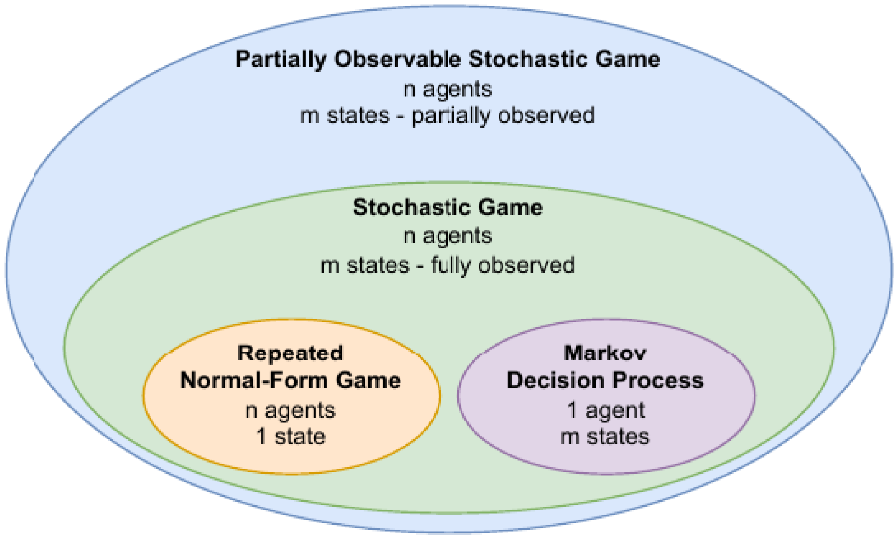


Figure 1: alt text

Normal-form game (game in strategic form) (Нормальная форма игры)

1. A finite set of N players.
2. Action set for the players: $\{\mathcal{A}_1, \mathcal{A}_2, \dots \mathcal{A}_N\}$
3. Reward functions for the k player: $R_n : \mathcal{A}_1 \times \mathcal{A}_2 \dots \times \mathcal{A}_N \to \mathbb{R}$ (can be represented as matrix, two players -> two matrices bn)

Types of games

example the Prisoner's Dilemma

Reward is closely related to preference $u_1(Fink, Quiet) > u_1(Quiet, Quiet) > u_1(Fink, Fink) > u_1(Q$

$$a_1 \preccurlyeq a_2 \Leftrightarrow u(a_1) \leq u(a_2)$$

	Head	*Tail*
Head	$1, -1$	$-1, \ \ 1$
Tail	$-1, \ \ 1$	$1, -1$

example **Figure 17.1** *Matching Pennies* (Example 17.1).

RL (this book)	GT	Description
environment	game	Model specifying the possible actions, observations, and rewards of agents, and the dynamics of how the state evolves over time and in response to actions.
agent	player	An entity which makes decisions. "Player" can also refer to a specific role in the game that is assumed by an agent, for example: • "row player" in a matrix game • "white player" in chess
reward	payoff, utility	Scalar value received by an agent/player after taking an action.
policy	strategy	Function used by an agent/player to assign probabilities to actions. In game theory, "(pure) strategy" is also sometimes used to refer to actions.
deterministic X	pure X	X assigns probability 1 to one option, for example: • deterministic policy • pure strategy • deterministic/pure Nash equilibrium
probabilistic X	mixed X	X assigns probabilities ≤ 1 to options, for example: • probabilistic policy • mixed strategy • probabilistic/mixed Nash equilibrium
joint X	X profile	X is a tuple, typically with one element for each agent/player, for example: • joint reward • deterministic joint policy • payoff profile • pure strategy profile

Figure 2: alt text

Normal-form games can be classified based on the relationship between the reward functions of agents:

- In a *zero-sum* game, the sum of the agents' rewards is always 0, i.e., $\sum_{i \in I} \mathcal{R}_i(a) = 0$ for all $a \in A$.[2] In zero-sum games with two agents, i and j,[3] one agent's reward function is simply the negative of the other agent's reward function, i.e., $\mathcal{R}_i = -\mathcal{R}_j$.
- In a *common-reward* game, all agents receive the same reward, i.e., $\mathcal{R}_i = \mathcal{R}_j$ for all $i, j \in I$.
- In a *general-sum* game, there are no restrictions on the relationship of reward functions.

Figure 3: img_4.png

		Suspect 2	
		Quiet	*Fink*
Suspect 1	*Quiet*	2, 2	0, 3
	Fink	3, 0	1, 1

Figure 13.1 The *Prisoner's Dilemma* (Example 12.1).

Figure 4: alt text

example 

	Stag	Hare
Stag	2, 2	0, 1
Hare	1, 0	1, 1

Figure 18.1 The *Stag Hunt* (Example 18.1) for the case of two hunters.

example Rock-scisor-paper game

example Hawk-dove games

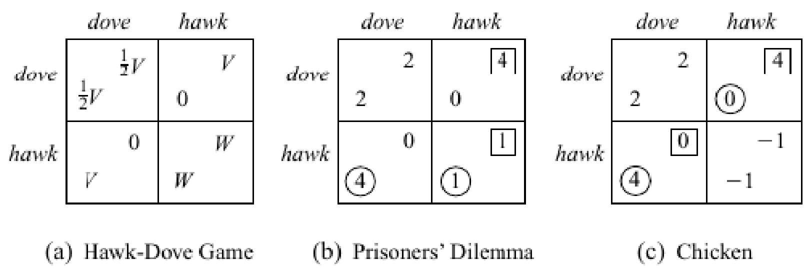


Figure 6.3 Hawk-Dove Games.

Figure 5: img_12.png

Mixed strategies

$$\pi = (\pi_1, ..\pi_i.., \pi_N) = (\pi_i, \pi_{-i})$$

Expected return for mixed strategies (utility)

$$J_n(\pi) = \mathbb{E}_\pi[R_n] = \sum_{(a_1,..,a_N)\in A_1\times\ldots\times A_N} \pi_1(a_1)\cdot\ldots\cdot\pi_n(a_n)\cdot R_n(a_1, ..., a_N) = \sum_{a_n\in A_n} \pi_1(a_n) \Bigg[\sum_{a_{-n}\in A_-}$$

For two players game reward vector will be equal $R_k((\pi_1, \pi_2)) = \pi_1 R_k \pi_2$

reaction curve

Spaces of mixed strategies

Payoff regions

cooperative payoff regions: $\pi(a_n, a_{-n})$ не факторизуемая, для $\forall\pi(a_1, ..., a_N)$:

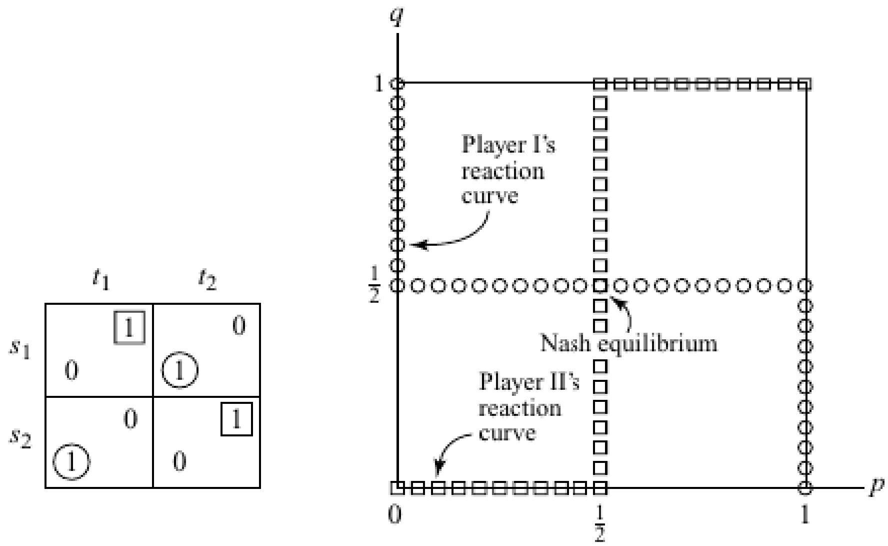


Figure 6: img_10.png

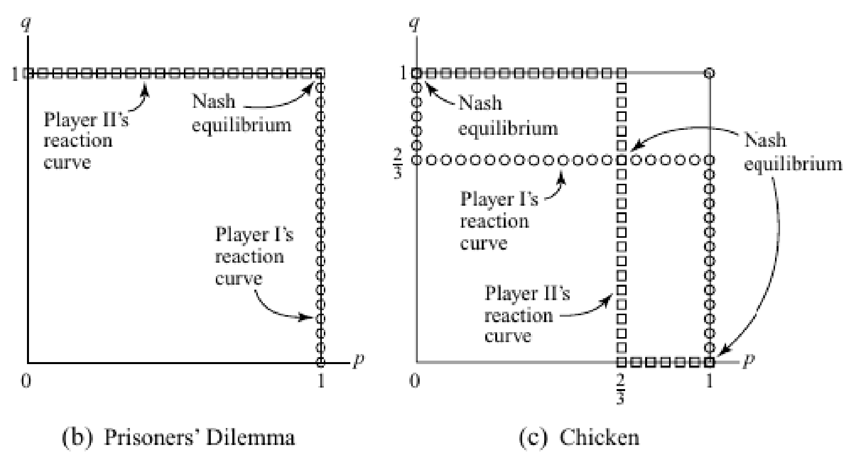


Figure 7: img_11.png

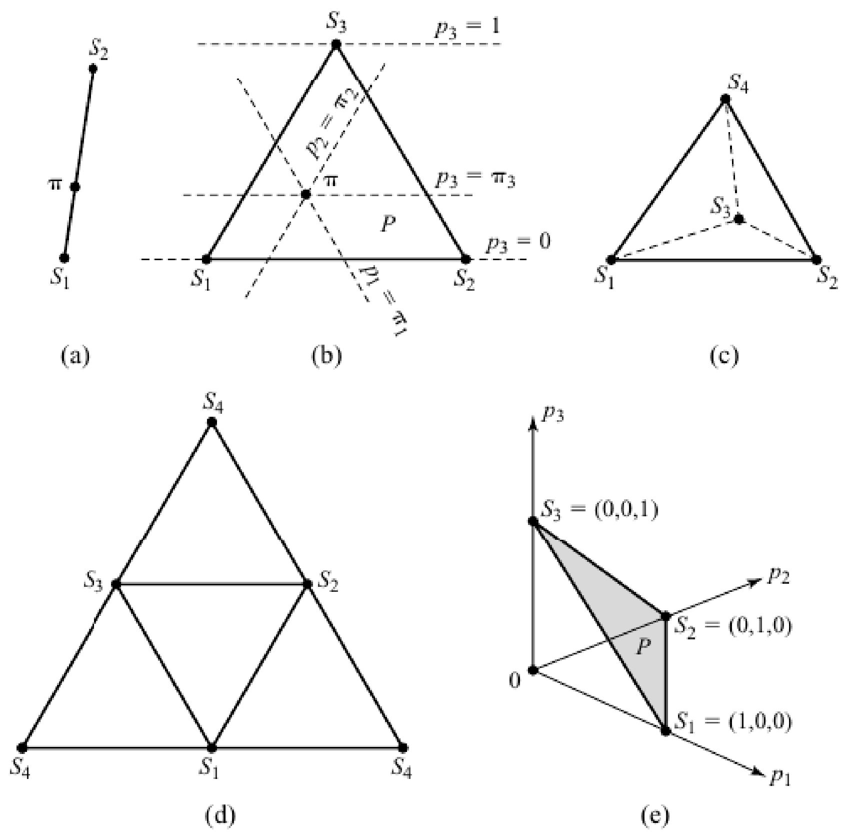
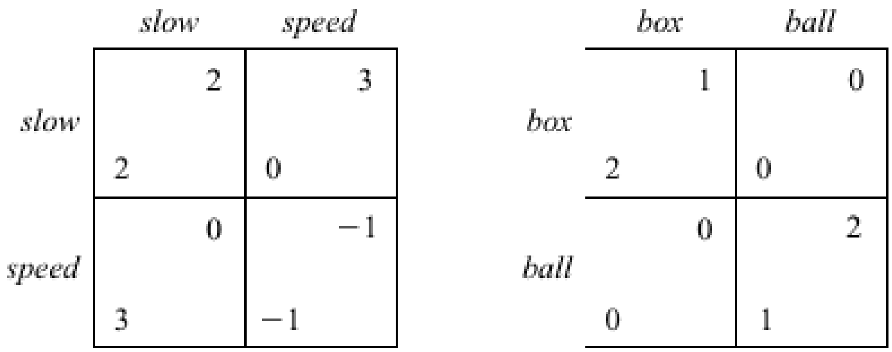


Figure 6.13 Spaces of mixed strategies. A contour labeled $p_i = \pi_i$ in Figure 6.13(b) consists of all points $p = p_1 s_1 + p_2 s_2 + p_3 s_3$ with $p_i = \pi_i$ and $p_1 + p_2 + p_3 = 1$. These contours are straight lines (Exercise 6.9.25). The faces of the tetrahedron of Figure 6.13(c) that meet at the vertex s_4 have been peeled away and the whole laid flat on the page to produce Figure 6.13(d). The point s_4 therefore appears three different times in the latter figure. One can similarly think of Figure 6.13(b) as the triangle P of Figure 6.13(e) laid flat on the page.

Figure 8: img_13.png



(a) Chicken (b) Battle of the Sexes

Figure 9: img_15.png

$$J_n(\pi) = \mathbb{E}_\pi[R_n] = \sum_{(a_1,..,a_N)\in A_1\times\ldots\times A_N} \pi(a_1, ..., a_N) \cdot R_n(a_1, ..., a_N)$$

следовательно convex function

noncooperative payoff regions:

Solution concepts

Nash equilibrium

$$\forall n R_n(\pi_n^*, \pi_{-n}^*) \geq R(\pi_n, \pi_{-n}^*)$$

Best responce function

set-valued function $B_i(a_{-i}) = \{a_i \text{ in } A_i : u_i(a_i, a_{-i}) \geq u_i(a_i', a_{-i}) \text{ for all } a_i' \text{ in } A_i\} :$

for mixed strategies $B_n(\pi_{-n}) = \arg\max_{\pi_n} J_n(\pi_n, \pi_{-n})$

Nash equlibrium in zero sum game = minimax (maximin)

a player has a profitable deviation if and only if they have a profitable deviation to a pure strategy can be established through the following proof.

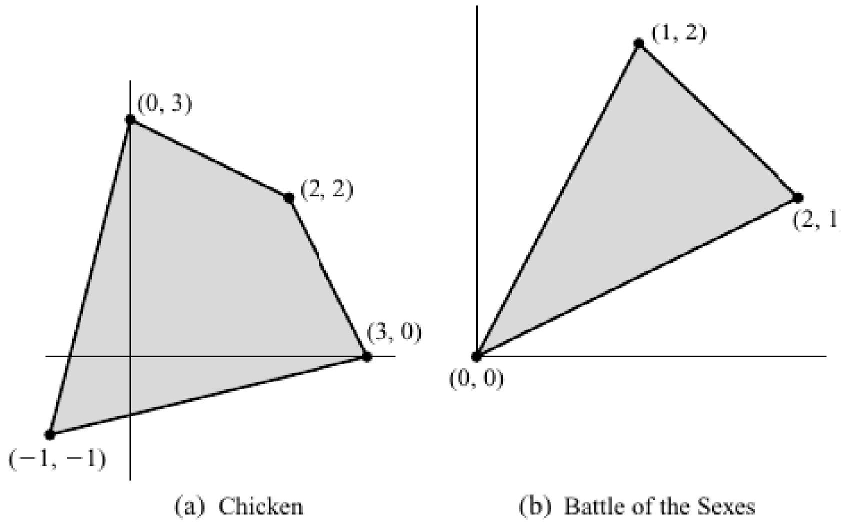


Figure 6.16 Cooperative payoff regions.

Figure 10: img_14.png

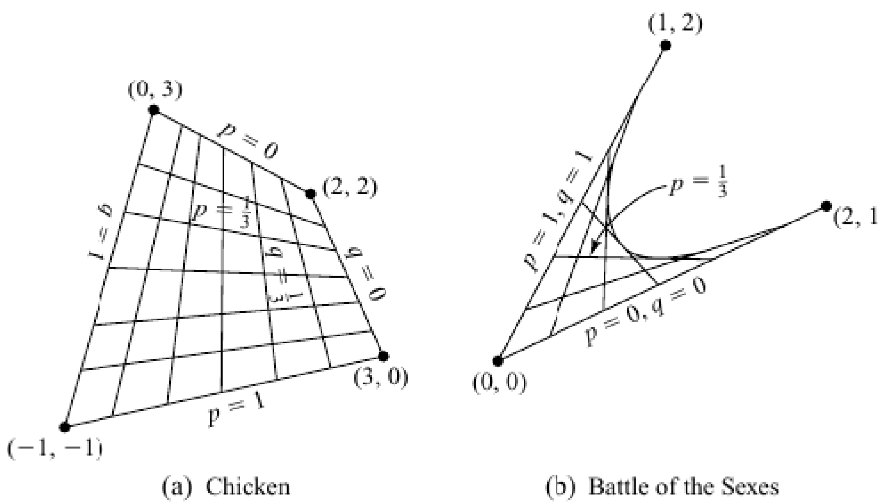


Figure 6.17 Noncooperative payoff regions.

Figure 11: img_16.png

▶ DEFINITION 21.1 (*Nash equilibrium of strategic game with ordinal preferences*) The action profile a^* in a strategic game with ordinal preferences is a **Nash equilibrium** if, for every player i and every action a_i of player i, a^* is at least as good according to player i's preferences as the action profile (a_i, a^*_{-i}) in which player i chooses a_i while every other player j chooses a^*_j. Equivalently, for every player i,

$$u_i(a^*) \geq u_i(a_i, a^*_{-i}) \text{ for every action } a_i \text{ of player } i, \tag{21.2}$$

where u_i is a payoff function that represents player i's preferences.

Figure 12: alt text

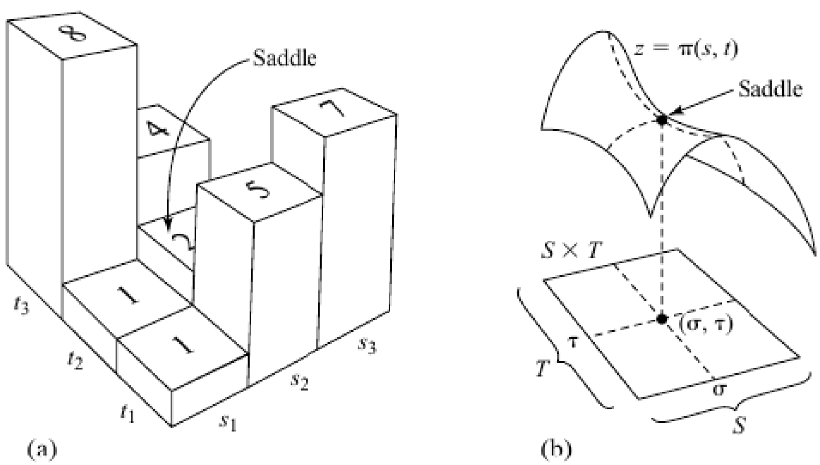


Figure 7.5 Saddle points.

Figure 13: img_17.png

Definitions

1. Mixed Strategy: A strategy where a player randomizes over two or more pure strategies.
2. Pure Strategy: A strategy where a player chooses one specific action with certainty.
3. Payoff: The outcome a player receives from a particular strategy profile.

Proof

1. If a player has a profitable deviation to a mixed strategy, then they have a profitable deviation to a pure strategy.

Assume player i is currently playing a mixed strategy σ_i in a game where the other players are playing strategies σ_{-i} .

Let $u_i(\sigma_i, \sigma_{-i})$ denote the payoff to player i when playing strategy σ_i against the strategies of the other players σ_{-i} .

If player i has a profitable deviation to another mixed strategy σ'_i , this means:

$u_i(\sigma'_i, \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$

Now, because mixed strategies are composed of probabilities over pure strategies, there exists at least one pure strategy s_i such that:

$u_i(s_i, \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$

This is due to the fact that if the mixed strategy σ'_i is better than σ_i , then at least one of the pure strategies in σ'_i must yield a higher payoff against σ_{-i} than σ_i . Thus, we conclude that player i has a profitable deviation to some pure strategy.

2. If a player has a profitable deviation to a pure strategy, then they have a profitable deviation to a mixed strategy.

Conversely, suppose player i can deviate to a pure strategy s_i' such that:

$u_i(s_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$

Now, consider the mixed strategy σ'_i that assigns probability 1 to playing s_i' (and probability 0 to all other strategies). Then:

$u_i(\sigma'_i, \sigma_{-i}) = u_i(s_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$

Thus, this mixed strategy σ'_i represents a profitable deviation as well.

proof

define $\varphi_{n,a_n}(\pi = \pi_n, \pi_{-n}) = \max\{0, J_n(a_n, \pi_{-n})\}$

it is one of the best responces regarding action a_n because a player has a profitable deviation if and only if he has a profitable deviation to a pure strategy

define $f : \Pi = \Pi_1 \times \cdots \times \Pi_N \to \Pi$

$$f_{n,a_n}(\pi) = \frac{\pi_n(a_n) + \varphi_{n,a_n}(\pi)}{\sum_{b_n \in A_N} \left[\pi_n(b_n) + \varphi_{n,b_n}(\pi)\right]}$$

f is continuous (each $\varphi_{n,an}$ is continuous)

Π is compact (each Π_n is compact)

Then by Brouwer's fixed point theorem f has at least one fix point

$\Rightarrow$ If π is Nash eq, then $\forall n \forall a_n \varphi_{n,an} = 0$, then π is fixed point

$\Leftarrow$ $\exists a_n \in support(n) = \{a > 0\}$ such that $R_n(a_n, \pi_{-n}) \leq R_n(\pi_n, \pi_{\ n})$ by linearity of expectaion

Then $\varphi_{n,an} = 0$ (*)

$f(\pi)_{n,a_n} = \pi_{n,a_n}$ (**) because π is fixed point

(*), (**) $\Rightarrow \sum_{b_n \in A_N} \left[\pi_n(b_n) + \varphi_{n,b_n}(\pi)\right] = 1 \Rightarrow \forall b_n \varphi_{n,b_n} = 0$ beacause $\varphi_{n,b_n} \geq 0$

then no player can improve his expected payoff by moving to a pure strategy

then π is a Nash equilibrium

Weak domination

ϵ-Nash equlibrium

Pareto optimality

treat □as vector indexed by players

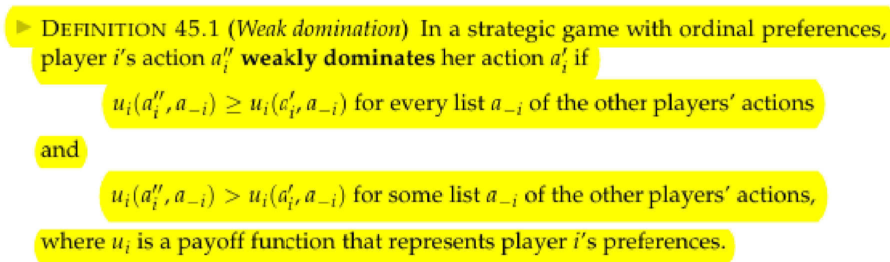
Extensive game (Развёрнутая форма игры)

a pure strategy of a player

is a collection of maps from all possible histories into available action

$I(x)$ - information set (generalization of idea of history) - waht player has when he is choosing his action

example



▶ DEFINITION 45.1 (*Weak domination*) In a strategic game with ordinal preferences, player i's action a_i'' **weakly dominates** her action a_i' if

$$u_i(a_i'', a_{-i}) \geq u_i(a_i', a_{-i}) \text{ for every list } a_{-i} \text{ of the other players' actions}$$

and

$$u_i(a_i'', a_{-i}) > u_i(a_i', a_{-i}) \text{ for some list } a_{-i} \text{ of the other players' actions,}$$

where u_i is a payoff function that represents player i's preferences.

Figure 14: img_2.png

The following two extensive form games are representations of the simultaneous-move matching pennies. The loops represent the information sets of the players who move at that stage. These are imperfect information games. These games represent exactly the same strategic situation: each player chooses his action not knowing the choice of his opponent.

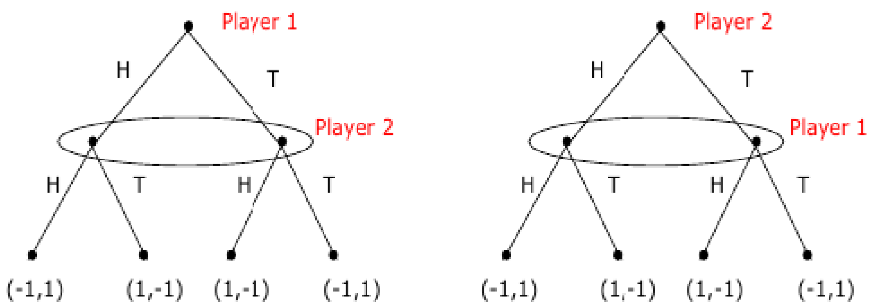


Figure 15: img_18.png

Deriving Normal Form from Extensive Form Games every extensive form game has a unique normal form representation

->

$x' \in I(x)$ means x' is indistinguishable from x

V_G - set of nodes of G

H - histories

subgame G' (of an extensive game G)

a single nnode of the G and its sucessors with property: $x_1 \in V_{G'}$ and $x_2 \in I(x_1)$, then $x_2 \in V_{G'}$ (If a node in a particular information

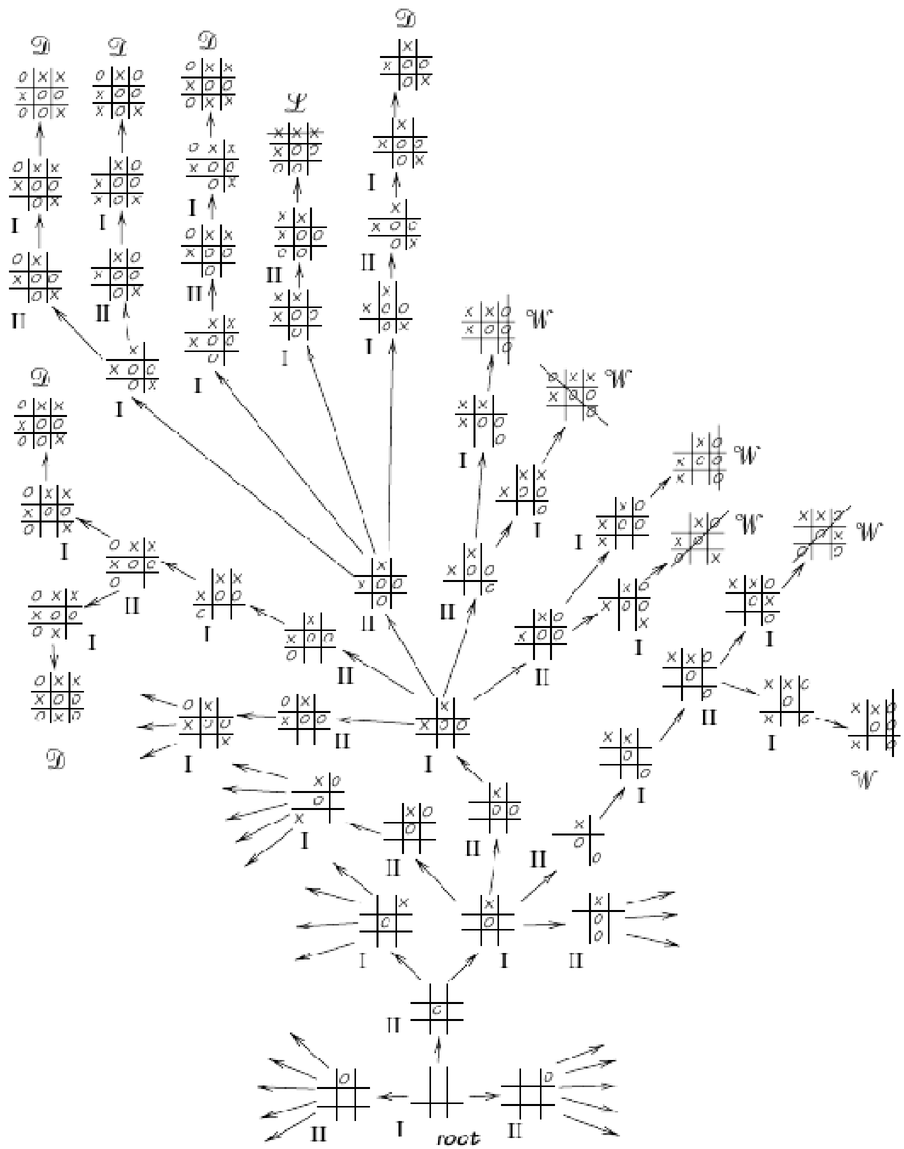


Figure 16: img_7.png

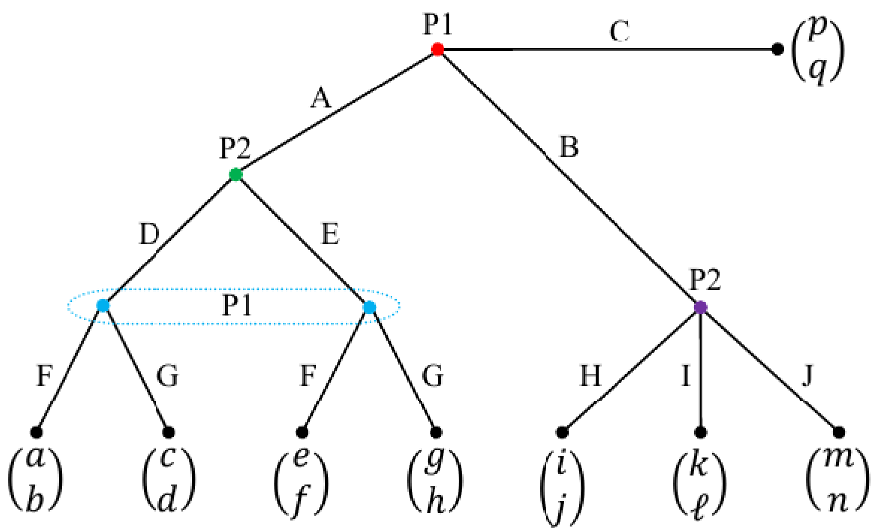


Figure 17: img_8.png

P1 \ P2	DH	DI	DJ	EH	EI	EJ
AF	a,b	a,b	a,b	e,f	e,f	e,f
AG	c,d	c,d	c,d	g,h	g,h	g,h
BF	i,j	k,ℓ	m,n	i,j	k,ℓ	m,n
BG	i,j	k,ℓ	m,n	i,j	k,ℓ	m,n
CF	p,q	p,q	p,q	p,q	p,q	p,q
CG	p,q	p,q	p,q	p,q	p,q	p,q

Figure 18: img_9.png

set is in the subgame then all members of that information set belong to the subgame.)

$H|_{G'} = \{h' = (a^t, \ldots, a^T) : (h, h') = (a^1, \ldots, a^T) \in H\}$

subgame perfect nash equlibrium (SPE) in game G

A strategy profile s^* s.t. for any subgame G' of G , $s^*|_{G'}$ (action profile implied by s in the subgame G') is a Nash equilibrium of G .

Backward induction refers to starting from the last subgames of a finite game, then finding the best response strategy profiles or the Nash equilibria in the subgames, then assigning these strategies profiles and the associated payoffs to be subgames, and moving successively towards the beginning of the game.

proof based on

One-stage Deviation Principle^ Informally, s is a subgame perfect equilibrium (SPE) if and only if no player i can gain by deviating from s in a single stage and conforming to s thereafter.

Repeated games

$r_i : A \to \mathbb{R}$ stage payoff for player i

$$R = \sum_t \gamma^t \cdot r(a_i^t, a_{-i}^t)$$

1. $T < \infty$

Theorem
For $T < \infty$ unique SPE $\pi^* = a^*$ for unique pure one stage equilibrium a^*

Example Finitely-Repeated Prisoners' Dilemma

	Cooperate	Defect
Cooperate	$1, 1$	$-1, 2$
Defect	$2, -1$	$0, 0$

Figure 19: img_19.png

In the last period,“defect” is a dominant strategy regardless of the history of the game. So the subgame starting at T has a dominant strategy equilibrium: (D, D).

Then move to stage $T-1$. By backward induction, we know that at T , no matter what, the play will be (D, D). Then given this, the subgame starting at $T-1$ (again regardless of history) also has a dominant strategy equilibrium.

With this argument, we have that there exists a unique SPE:$ (D, D)$ at each date.

2. $T = \infty$

$$R = (1-\gamma)\sum_t \gamma^t \cdot r(a_i^t, a_{-i}^t)$$

Trigger strategy

A trigger strategy essentially threatens other players with a “worse,” punishment, action if they deviate from an implicitly agreed action profile.

Example Infinitely-Repeated Prisoners’ Dilemma

SPE - use action C unless someone not using first time D. In that case alse use D

From moment t use action C

$R = (1-\gamma)(1+\gamma+\gamma^2+...) = 1$

From moment t deviate and use action D

$R = (1-\gamma)(2+0+0+...) = 2\cdot(1-\gamma)$

cooperation better if $2\cdot(1-\gamma) \geq 1$

Folk Theorem

Learning solution in game

1. Ficticious play (store as memory opponents moves statistic)
2. Evolution in game:

a. The replicator dynamic (continuous time) - can be used to learn solution

b. The idea of an Evolutionary Stable (only definition of solution) Strategy or ESS.

Evolutionary Stable Strategy

symmetric two-player game

current strategy π

small fraction (ϵ) select different strategy π

The expected payoff of a mutant is: $(1-\epsilon)r(\pi, \pi^*) + \epsilon r(\pi, \pi)$

The expected payoff of no-mutant player is: $(1-\epsilon)r(\pi^*, \pi^*) + \epsilon r(\pi^*, \pi)$

evolutionary stable strategy (1)

$\pi^* \in \Delta : \exists \epsilon'$ such that for any π for any $\epsilon < \epsilon'$

$$r(\pi^*, (1-\epsilon)\pi^* + \epsilon\pi) > r(\pi, (1-\epsilon)\pi^* + \epsilon\pi)$$

evolutionary stable strategy (2)

$\pi^* \in \Delta\ \forall \pi \neq \pi^* :$ 1. $r(\pi^*, \pi^*) \geq r(\pi, \pi^*)$ 2. if $r(\pi^*, \pi^*) = r(\pi, \pi^*)$ then $r(\pi, \pi) < r(\pi^*, \pi)$

definitions (1) and (2) are equal

based on following property

$$r(\pi^*, (1-\epsilon)\pi^* + \epsilon\pi) = (1-\epsilon)r(\pi^*, \pi^*) + \epsilon r(\pi^*, \pi)$$

because $r(\pi_1, \pi_2) = J(\pi_1, \pi_2) = \mathbb{E}_{(a_1, a_2)}[R(a_1, a_2)]$

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